IIT BOMBAY TECHFEST 2026–27 | ROUND 1 DELIVERABLE

NammaSpace

Interactive 3D Digital Twin & Real-Time Spatial Navigation Engine

Challenge: NammaSpace
Round: 1 (3D Reconstruction & Roaming)
Status: Live Production Deployed

This official submission document provides direct access to the live, zero-install WebGL viewer and open-source code repository for the **NammaSpace** spatial digital twin challenge. All assets have been compiled and hosted for instant browser testing by the IIT Bombay Techfest evaluation panel.

🔗 OFFICIAL PROJECT ACCESS LINKS

🌐 1. Live Web Application (Zero-Install WebGL Viewer)

https://nammaspace-digital-twin.vercel.app/

Runs locked at **60+ FPS** in any modern desktop or mobile browser (Chrome, Edge, Safari, Brave). Features first-person exploration, a **Triple-Environment Showcase (Studio 1.13M, Atrium 1.73M, Courtyard 3.14M — Total 6.0M+ splats)**, Google Maps indoor search & 3D pins, obstacle-avoiding A* walking pathfinding, and a synchronized architectural 2D floor radar.

💻 2. Open-Source Source Code & Pipeline Repository

https://github.com/Kaish-sheikh/nammaspace-digital-twin

Complete codebase containing the Three.js WebGL engine, 3D Gaussian Splatting scene loaders, Catmull-Rom pathfinding algorithms, Jacobi ground-plane eigendecomposition leveling, monocular capture Standard Operating Procedure (SOP), and sample trained splat binaries.

🎮 QUICK EVALUATION GUIDE FOR JUDGES

🕹️ Locomotion & Controls

Use `W / A / S / D` to walk and `SHIFT` to sprint. Move your mouse to look around in first-person view. Middle-click or `ESC` frees the cursor; clicking the 3D scene smoothly re-engages walk mode.

🔄 Triple-Environment Switcher

Toggle between **Studio (1.13M)**, **Atrium (1.73M)**, and **Courtyard (3.14M)** via the top header. All models are mathematically pre-leveled to ground plane Y=0.

🔍 Google Maps Indoor Search & 3D Pins

Type in the search bar to filter destinations; press `Enter` to auto-orient the camera. Toggle `Pins: OFF / ON` to reveal 3D teardrop markers with floor target rings.

🧭 Obstacle-Aware A* Waypoint Pathfinding

Select any waypoint to project an animated neon dashed walking path across the floor avoiding room obstacles, with live distance and travel time estimation.

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